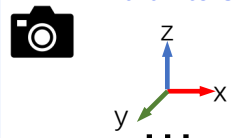


a Feed-Forward Gaussian Prediction

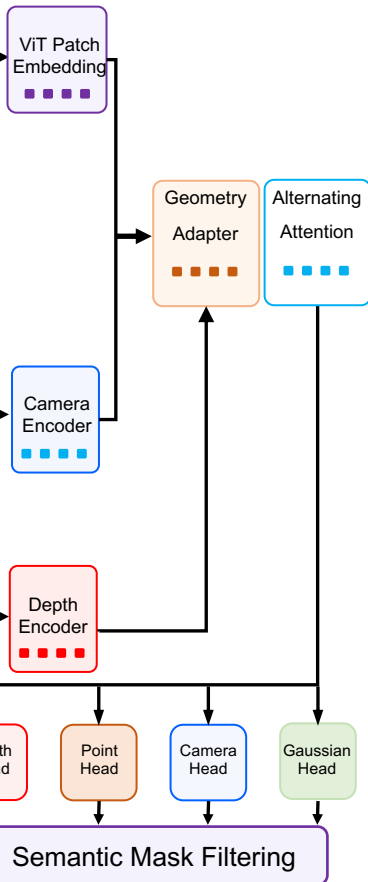
RGB Views



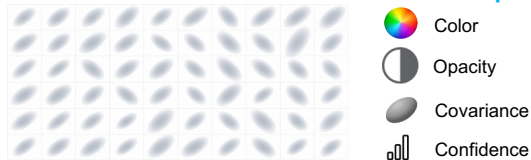
Camera Parameters



Depth Maps

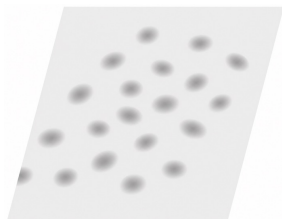


Dense Pixel-wise Gaussian Map



b Coordinate Alignment

Predicted Space (Normalized Coordinate)

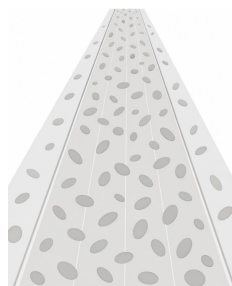


Umeyama Alignment

Gaussian Transformation



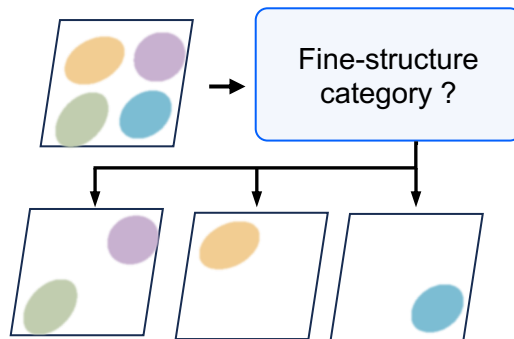
GT World Space (World Coordinate)



- Depth Alignment
- Scale Alignment
- Rotation Alignment

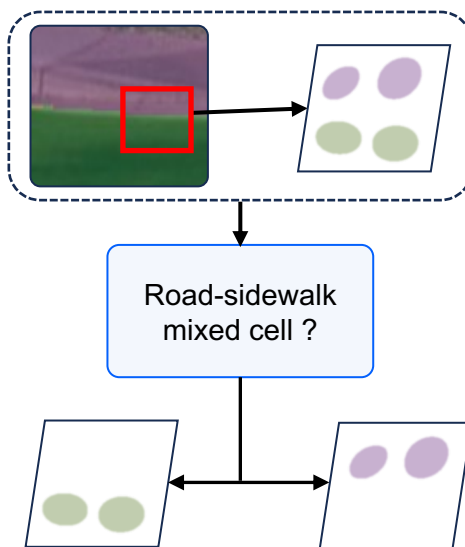
c Road-Structure-Aware Grouping

1 Category-based Grouping



2 Road-Sidewalk Junction Protection

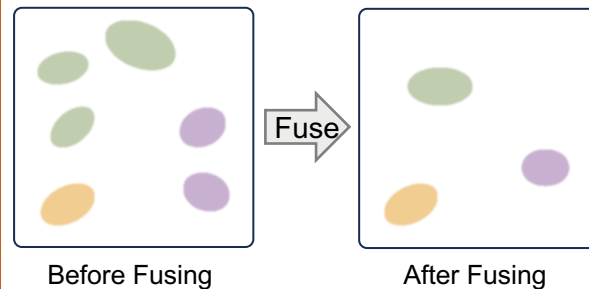
Dangerous Elevation Difference



Avoid Cross-Category Fusing to Preserve More Details

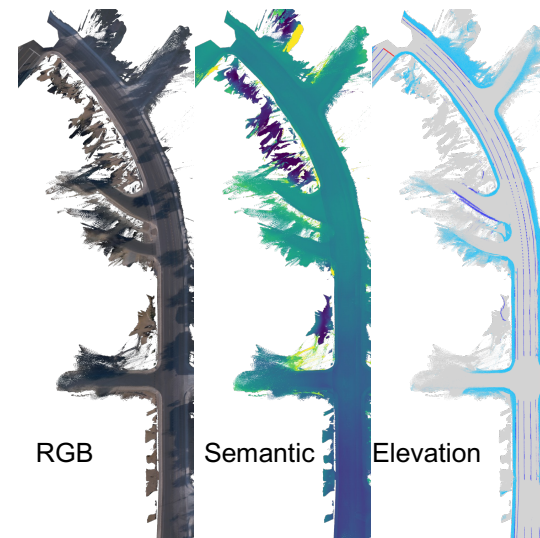
d Grid-based Gaussian Fusion

For Each Group within a Grid Cell



e Applications

Supported BEV Applications



Novel View Synthesis

